Partial Fractions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distinct linear factors**

Express $(5x + 7)/((x + 1)(x + 2))$ in partial fractions.

Write $(5x + 7)/((x + 1)(x + 2)) = A/(x + 1) + B/(x + 2)$. Multiply through: $5x + 7 = A(x + 2) + B(x + 1)$. At $x = -1$: $2 = A(1)$, so $A = 2$. At $x = -2$: $-3 = B(-1)$, so $B = 3$. Result: $2/(x + 1) + 3/(x + 2)$.

2 **[Foundational]** • **Target: Repeated linear factors**

Express $(4x + 1)/(x - 1)^2$ in partial fractions of the form $A/(x - 1) + B/(x - 1)^2$.

$4x + 1 = A(x - 1) + B$. At $x = 1$: $5 = B$. Comparing coefficients of x: $A = 4$. Result: $4/(x - 1) + 5/(x - 1)^2$.

3 **[Foundational]** • **Target: Applying partial fractions to integration**

Use partial fractions to evaluate $\int (5x + 7)/((x + 1)(x + 2))\,dx$, using the decomposition found earlier in this worksheet.

From earlier: $(5x + 7)/((x + 1)(x + 2)) = 2/(x + 1) + 3/(x + 2)$. So $\int (5x + 7)/((x + 1)(x + 2))\,dx = 2\ln|x + 1| + 3\ln|x + 2| + C$.

4 **[Foundational]** • **Target: Applying partial fractions to a binomial series expansion**

Express $1/((1 - x)(1 - 2x))$ in partial fractions, then use the result to write the first three terms of its series expansion in ascending powers of x.

$1/((1 - x)(1 - 2x)) = A/(1 - x) + B/(1 - 2x)$. Then $1 = A(1 - 2x) + B(1 - x)$. At $x = 1$: $1 = A(-1)$, so $A = -1$. At $x = 1/2$: $1 = B(1/2)$, so $B = 2$. Result: $-1/(1 - x) + 2/(1 - 2x)$. Expanding each geometric series: $-1(1 + x + x^2 + \cdots) + 2(1 + 2x + 4x^2 + \cdots) = (-1 + 2) + (-1 + 4)x + (-1 + 8)x^2 + \cdots = 1 + 3x + 7x^2 + \cdots$.

5 **[Foundational]** • **Target: Combining terms before decomposing**

Express $(x^2 + x + 1)/(x(x + 1))$ in partial fractions. (Hint: the degree of the numerator equals the degree of the denominator, so first perform polynomial division.)

Since numerator and denominator are both degree 2, divide first: $(x^2 + x + 1)/(x^2 + x) = 1 + 1/(x^2 + x) = 1 + 1/(x(x + 1))$. Decompose $1/(x(x + 1)) = A/x + B/(x + 1)$: $1 = A(x + 1) + Bx$. At $x = 0$: $A = 1$. At $x = -1$: $B = -1$. Result: $1 + 1/x - 1/(x + 1)$.

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6 [Foundational] • Target: Applying partial fractions to a rate problem

A population model gives the rate of change $dP/dt = 1/(P(10 - P))$. Express $1/(P(10 - P))$ in partial fractions of the form $A/P + B/(10 - P)$.

$1 = A(10 - P) + BP$. At $P = 0$: $1 = 10A$, so $A = 1/10$. At $P = 10$: $1 = 10B$, so $B = 1/10$. Result: $1/(10P) + 1/(10(10 - P))$ — a decomposition commonly used to solve logistic-growth differential equations by separation of variables.

7 [Foundational] • Target: Partial fractions with an improper fraction and repeated factor

Express $x^3/(x - 1)^2$ in partial fractions, first performing polynomial division since the numerator has higher degree than the denominator.

Divide x^3 by $(x - 1)^2 = x^2 - 2x + 1$: $x^3 \div (x^2 - 2x + 1) = x + 2$ remainder $3x - 2$. So $x^3/(x - 1)^2 = x + 2 + (3x - 2)/(x - 1)^2$. Decompose the remainder: $(3x - 2)/(x - 1)^2 = A/(x - 1) + B/(x - 1)^2$, giving $3x - 2 = A(x - 1) + B$. At $x = 1$: $1 = B$. Comparing x coefficients: $A = 3$. Result: $x + 2 + 3/(x - 1) + 1/(x - 1)^2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using partial fractions to sum a series

Express $1/(n(n + 1))$ in partial fractions, and use the result to find $1/(1 \cdot 2) + 1/(2 \cdot 3) + 1/(3 \cdot 4) + 1/(4 \cdot 5)$ by telescoping.

$1/(n(n + 1)) = 1/n - 1/(n + 1)$ (standard decomposition with $A = 1, B = -1$). Summing from $n = 1$ to 4: $(1 - 1/2) + (1/2 - 1/3) + (1/3 - 1/4) + (1/4 - 1/5) = 1 - 1/5 = 4/5$, since all the middle terms cancel (telescope).

9 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

Use partial fractions to find the Maclaurin (power series) expansion of $1/((1 - x)(2 - x))$ up to the term in x^2, first decomposing into the form $A/(1 - x) + B/(2 - x)$.

$1 = A(2 - x) + B(1 - x)$. At $x = 1$: $1 = A(1)$, so $A = 1$. At $x = 2$: $1 = B(-1)$, so $B = -1$. Result: $1/(1 - x) - 1/(2 - x) = 1/(1 - x) - 1/2 \cdot 1/(1 - x/2)$. Expanding each as a geometric series:

$(1 + x + x^2 + \cdots) - 1/2(1 + x/2 + x^2/4 + \cdots) = (1 - 1/2) + (1 - 1/4)x + (1 - 1/8)x^2 + \cdots = 1/2 + 3/4x + 7/8x^2 + \cdots$.

10 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

Express $(4x^2 - 5x - 15)/((x + 1)(x - 2)(x + 3))$ in partial fractions, using the cover-up method (evaluating the numerator at each root while covering the matching factor in the denominator).

Write $(4x^2 - 5x - 15)/((x + 1)(x - 2)(x + 3)) = A/(x + 1) + B/(x - 2) + C/(x + 3)$. Using the cover-up method:

$A = (4(-1)^2 - 5(-1) - 15)/((-1 - 2)(-1 + 3)) = (4 + 5 - 15)/((-3)(2)) = -6/(-6) = 1$. $B = (4(2)^2 - 5(2) - 15)/((2 + 1)(2 + 3)) = (16 - 10 - 15)/((3)(5)) = -9/15 = -3/5$.

$C = (4(-3)^2 - 5(-3) - 15)/((-3 + 1)(-3 - 2)) = (36 + 15 - 15)/((-2)(-5)) = 36/10 = 18/5$. Result: $1/(x + 1) - 3/(5(x - 2)) + 18/(5(x + 3))$.

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11 **[Intermediate]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure**

A chemical reaction's rate is modeled by $dC/dt = 1/((C+2)(C-1))$, where C is a concentration. Express $1/((C+2)(C-1))$ in partial fractions of the form $A/(C+2) + B/(C-1)$, and then use the result to write the general antiderivative $\int 1/((C+2)(C-1))\,dC$ in terms of natural logarithms.

$1 = A(C-1) + B(C+2)$. At $C = 1$: $1 = B(3)$, so $B = 1/3$. At $C = -2$: $1 = A(-3)$, so $A = -1/3$. Result: $-1/3/(C+2) + 1/3/(C-1)$. Integrating: $\int 1/((C+2)(C-1))\,dC = -1/3\ln|C+2| + 1/3\ln|C-1| + K = 1/3\ln|(C-1)/(C+2)| + K$.

12 **[Intermediate]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure**

Use the decomposition $1/(3x) + 8/(3(x-3))$ found earlier in this worksheet for $(3x-1)/(x(x-3))$ to evaluate $\int_4^5 (3x-1)/(x(x-3))\,dx$, giving your answer as a single logarithm expression.

$\int (3x-1)/(x(x-3))\,dx = 1/3\ln|x| + 8/3\ln|x-3| + C$. Evaluating from 4 to 5:

$[1/3\ln 5 + 8/3\ln 2] - [1/3\ln 4 + 8/3\ln 1] = 1/3\ln 5 + 8/3\ln 2 - 1/3\ln 4 = 1/3\ln(5/4) + 8/3\ln 2$.

13 **[Intermediate]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure**

This question has three parts. (a) Factor $x^2 - 4$ and hence express $(x+8)/(x^2-4)$ in partial fractions. (b) Use your result from part (a) to write down $\int (x+8)/(x^2-4)\,dx$. (c) Hence evaluate $\int_3^5 (x+8)/(x^2-4)\,dx$, giving your answer as a single expression involving natural logarithms.

(a) Factor $x^2 - 4 = (x-2)(x+2)$, so $(x+8)/((x-2)(x+2)) = A/(x-2) + B/(x+2)$. Then $x + 8 = A(x+2) + B(x-2)$. At $x = 2$: $10 = 4A$, so $A = 5/2$. At $x = -2$: $6 = -4B$, so $B = -3/2$. Result: $5/2/(x-2) - 3/2/(x+2)$. (b) $\int (x+8)/(x^2-4)\,dx = 5/2\ln|x-2| - 3/2\ln|x+2| + C$. (c) Evaluating from 3 to 5: $[5/2\ln 3 - 3/2\ln 7] - [5/2\ln 1 - 3/2\ln 5] = 5/2\ln 3 - 3/2\ln 7 + 3/2\ln 5$ (since $\ln 1 = 0$).

14 **[Intermediate]** • **Target: Combining partial fractions back into a single fraction**

Combine $2/(x+1) + 3/(x+2)$ into a single fraction.

The common denominator is $(x+1)(x+2)$. So $2/(x+1) + 3/(x+2) = (2(x+2) + 3(x+1))/((x+1)(x+2))$. Expanding the numerator: $2(x+2) + 3(x+1) = 2x + 4 + 3x + 3 = 5x + 7$. Result: $(5x+7)/((x+1)(x+2))$ — confirming the original expression from the first question of this worksheet.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Solving a separable differential equation using partial fractions • Cross-Lesson: 1.01 Proof by Mathematical Induction

Solve the differential equation $dP/dt = P(10 - P)/100$, given that $P = 5$ when $t = 0$. Use the partial fraction decomposition $1/(P(10 - P)) = 1/(10P) + 1/(10(10 - P))$ found earlier in this worksheet to integrate, and give P explicitly as a function of t.

Separating variables: $dP/(P(10 - P)) = dt/100$. Using the decomposition, the left side integrates to $(1/10)\ln|P| - (1/10)\ln|10 - P| = (1/10)\ln|P/(10 - P)|$. So $(1/10)\ln|P/(10 - P)| = t/100 + C_1$. Multiplying by 10: $\ln|P/(10 - P)| = t/10 + C_2$. Exponentiating: $P/(10 - P) = ke^{t/10}$ for some constant k. At $t = 0$, $P = 5$, so $5/5 = 1 = k$. So $P/(10 - P) = e^{t/10}$, which rearranges to $P = 10e^{t/10}/(1 + e^{t/10})$.

16 [Challenging] • Target: Explaining when polynomial division is required • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Explain why polynomial division must be performed before applying partial fraction decomposition to $x^3/(x - 1)^2$, but not to $(3x + 1)/(x - 1)^2$.

The standard partial fraction method requires the numerator's degree to be strictly less than the denominator's degree. In $x^3/(x - 1)^2$, the numerator has degree 3 while the denominator $(x - 1)^2$ has degree 2, so the fraction is improper and must first be reduced by polynomial division into a polynomial plus a proper fraction before decomposing. In $(3x + 1)/(x - 1)^2$, the numerator has degree 1, which is already less than the denominator's degree 2, so the fraction is already proper and can be decomposed dir.

17 [Challenging] • Target: Verifying a decomposition by substitution • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Verify the decomposition $(5x + 7)/((x + 1)(x + 2)) = 2/(x + 1) + 3/(x + 2)$ found earlier in this worksheet by substituting $x = 0$ into both sides.

LHS: $(5(0) + 7)/((0 + 1)(0 + 2)) = 7/2$. RHS: $2/(0 + 1) + 3/(0 + 2) = 2 + 3/2 = 7/2$. Since both sides equal $7/2$, the decomposition checks out at $x = 0$.

18 [Challenging] • Target: Choosing efficient substitution values • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

When decomposing $(2x + 3)/((x - 4)(x + 5))$ using the equation $2x + 3 = A(x + 5) + B(x - 4)$, which two values of x should be substituted to find A and B most efficiently, and why?

Substitute $x = 4$ and $x = -5$ — the roots of the denominator's factors — since each choice makes one of the two terms on the right vanish, isolating A or B directly without solving simultaneous equations. At $x = 4$: $11 = 9A$, so $A = 11/9$. At $x = -5$: $-7 = -9B$, so $B = 7/9$.

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19 [Challenging] • Target: Distinct linear factors • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.

1. Identify $P(x) = \frac{2}{x}$ and $Q(x) = 4x$. 2. Calculate the integrating factor $\mu(x)$: $\mu(x) = \exp\left(\int \frac{2}{x}\,dx\right) = e^{2\ln x} = e^{\ln(x^2)} = x^2$. 3. Multiply through by x^2: $x^2\frac{dy}{dx} + 2xy = 4x^3 \Rightarrow \frac{d}{dx}[x^2y] = 4x^3$. 4. Integrate both sides: $x^2y = \int 4x^3\,dx = x^4 + C$. 5. Apply $y(1) = 2$: $(1)^2(2) = 1^4 + C \Rightarrow 2 = 1 + C \Rightarrow C = 1$. 6. Explicit solution: $y = \frac{x^4+1}{x^2} = x^2 + \frac{1}{x^2}$.

20 [Challenging] • Target: Repeated linear factors • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

1. Write the characteristic equation: $m^2 - 5m + 6 = 0 \Rightarrow (m-2)(m-3) = 0 \Rightarrow m_1 = 2,\ m_2 = 3$. 2. General solution: $y(x) = Ae^{2x} + Be^{3x}$. 3. Derivative: $y'(x) = 2Ae^{2x} + 3Be^{3x}$. 4. Apply initial conditions: $y(0) = 1 \Rightarrow A + B = 1$. $y'(0) = 3 \Rightarrow 2A + 3B = 3$. Multiply first equation by 2: $2A + 2B = 2$. Subtract from second: $B = 1$. Then $A = 1 - 1 = 0$. Particular Solution: $y(x) = e^{3x}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying the correct form of a decomposition

Which of the following is the correct form for the partial fraction decomposition of $(3x + 1)/((x - 2)^2(x + 1))$?

a. A/(x-2)+B/(x-2)^2+C/(x+1) ✓

b. A/(x-2)+B/(x+1)

c. A/((x-2)^2)+B/(x+1)

d. (Ax+B)/(x-2)^2+C/(x+1)

A repeated linear factor $(x - 2)^2$ requires two separate terms, $A/(x - 2)$ and $B/(x - 2)^2$, while the distinct linear factor $(x + 1)$ contributes one term, $C/(x + 1)$. The second option omits the repeated-factor structure, the third omits the lower-power term, and the fourth incorrectly uses a linear numerator over a repeated linear factor (that form is only needed for irreducible quadratic factors).

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22 [Foundational] • Target: Repeated linear factors

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

a. -3 ✓

b. 1

c. 3

d. 24

Step 1: Determine the polynomial quotient A. The degree of the numerator is 3 and the degree of the denominator is $1 + 2 = 3$. Expanding the denominator gives:
$(x+1)(x-2)^2 = (x+1)(x^2 - 4x + 4) = x^3 - 3x^2 + 4$ Since the leading coefficient of both the numerator and the denominator is 1, polynomial division yields a constant quotient: $A = \frac{1}{1} = 1$ Step 2: Set up the algebraic identity. Multiplying both sides of the partial fraction expansion by $(x+1)(x-2)^2$.

23 [Foundational] • Target: Irreducible quadratic factors

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$ ✓

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

Applying Techniques of Integration principles, $x = (x-1) + 1 \Rightarrow A = 1, B = 1$. Therefore, the verified correct option is $\frac{1}{x-1} + \frac{1}{(x-1)^2}$.

24 [Foundational] • Target: Applying partial fractions to integration

The rational function f is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x-2)(x^2+1)}$, where k is a real constant. Given that the series expansion of $f(x)$ in ascending powers of x, valid for sufficiently small $|x|$, has no term in x^2, find the value of k.

a. $\frac{25}{4}$ ✓

b. $\frac{17}{4}$

c. $-\frac{25}{4}$

d. $\frac{115}{12}$

To find the series expansion of $f(x)$, we first decompose $f(x)$ into partial fractions. **Step 1: Algebraic Division** The degree of the numerator is 4 and the degree of the denominator is $1 + 2 = 3$. Because the numerator has a higher degree than the denominator, $f(x)$ is an improper algebraic fraction and polynomial division must be carried out first. Expanding the denominator:
$(x-2)(x^2+1) = x^3 - 2x^2 + x - 2$. Dividing $2x^4 - 3x^3 + kx^2 - 8x + 3$ by $x^3 - 2x^2 + x - 2$.

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25 [Foundational] • Target: Applying partial fractions to a binomial series expansion

The first three terms of the binomial expansion of $(1 + kx)^n$ are $1 + 6x + 12x^2$. Find the value of k.

a. **2** ✓

b. -2

c. 3

d. -3

The binomial expansion of $(1 + x)^n$ is given by $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$. For the given expansion $(1 + kx)^n$, we can substitute $y = kx$ to get $(1 + y)^n$. The first three terms of the expansion are $1 + ny + \frac{n(n-1)}{2!}y^2 + \ldots$. Substituting $y = kx$ back, we have: $1 + n(kx) + \frac{n(n-1)}{2!}(kx)^2 + \ldots$ $1 + nkx + \frac{n(n-1)}{2!}k^2x^2 + \ldots$ We are given that the first three terms are $1 + 6x + 12x^2$. Comparing the coefficients: 1. Coefficient of x: $nk = 6$ 2. Coefficient of.

26 [Foundational] • Target: Combining terms before decomposing

Consider the quadratic function $f(x) = x^2 - 4x + 3$ and the linear function $g(x) = x - 1$. The points of intersection are given by their solutions. What is the y-coordinate of the point of intersection where x = 1?

a. **0** ✓

b. 1

c. -1

d. 2

To find the points of intersection, we set $f(x) = g(x)$: $x^2 - 4x + 3 = x - 1$. $x^2 - 5x + 4 = 0$. $(x - 1)(x - 4) = 0$. The x-coordinates of the intersection points are $x = 1$ and $x = 4$. We are asked for the y-coordinate when $x = 1$. We can use either function to find this y-coordinate. Using $g(x) = x - 1$, when $x = 1$, $g(1) = 1 - 1 = 0$. Therefore, the y-coordinate is 0.

27 [Foundational] • Target: Applying partial fractions to a rate problem

The graph of the quadratic function $f(x) = x^2 - 5x + 6$ is shown below. The graph intersects the x-axis at points A and B. Find the sum of the x-coordinates of points A and B.

a. **5** ✓

b. -5

c. 6

d. -6

The x-coordinates of points A and B are the roots of the quadratic equation $f(x) = 0$, which is $x^2 - 5x + 6 = 0$. For a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this equation, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $-\frac{-5}{1} = 5$.

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28 [Foundational] • Target: Partial fractions with three distinct linear factors

Generalised binomial: $\sqrt{1-x} = \sum_{n=0}^{\infty} \binom{1/2}{n}(-1)^n x^n$. First three terms:

a. $1 - x - x^2$

b. $1 + x/2 - x^2/8$

c. $1 - x/2 - x^2/8$ ✓

d. $1 - x/2$

Applying Convergence Tests and Power Series principles, $1 - x/2 - x^2/8 - \cdots$ Therefore, the verified correct option is $1 - x/2 - x^2/8$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Partial fractions with an improper fraction and repeated factor

Find $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ using partial fractions:

a. 1

b. 1/2

c. 2/3

d. 3/4 ✓

$\frac{1}{n(n+2)} = \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2})$; telescopes to $\frac{1}{2}(1 + 1/2) = 3/4$.

30 [Intermediate] • Target: Using partial fractions to sum a series

Partial fractions: if denominator has irreducible quadratic $x^2 + bx + c$, the numerator is:

a. A

b. $A/(x^2 + bx + c)$

c. $Ax^2 + B$

d. $Ax + B$ ✓

Linear numerator $Ax + B$ over irreducible quadratic.

31 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

Quadratic factor in partial fractions: $\frac{1}{x^2+4}$ gives integral:

a. $\frac{1}{2}\arctan(x/2) + C$ ✓

b. $\arctan x + C$

c. $\ln(x^2 + 4) + C$

d. $1/(2x + 4) + C$

$\int 1/(x^2 + 4)dx = \frac{1}{2}\arctan(x/2) + C$.

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32 [Intermediate] • Target: Combining partial fractions back into a single fraction

The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = ax^2 + bx + c$. Given that the vertex of the parabola is at $(-1, 8)$ and the graph passes through the point $(1, 0)$, find the values of a and b.

a. **a = -2, b = -4** ✓

b. a = -2, b = 4

c. a = 2, b = 4

d. a = 2, b = -4

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ is at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. We are given that the vertex is at $(-1, 8)$. Therefore, $-\frac{b}{2a} = -1$, which implies $b = 2a$. We are also given that the vertex is at $y = 8$, so $f(-1) = 8$. Substituting $x = -1$ into the function: $a(-1)^2 + b(-1) + c = 8 \Rightarrow a - b + c = 8$. Since $b = 2a$, we have $a - 2a + c = 8 \Rightarrow -a + c = 8$. The graph also passes through the point $(1, 0)$. So, $f(1) = 0$. Subst.

33 [Intermediate] • Target: Identifying the correct form of a decomposition

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, 1, -4)$, $B(1, 4, -2)$, and $C(-1, -3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 6

b. 9

c. 7

d. **14** ✓

Working through this vectors step: $\vec{AB} = (-1; 3; 2),\ \vec{AC} = (-3; -4; 8),\ \mathbf{n} = (32; 2; 13),\ d = \mathbf{n} \cdot A = 14$

34 [Intermediate] • Target: Solving a separable differential equation using partial fractions

A sphere's radius increases at $\frac{dr}{dt} = 3$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 5$ cm.

a. 320π cm^3/s

b. 310π cm^3/s

c. 300π cm^3/s ✓

d. 280π cm^3/s

Working through this calculus step: $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(5)^2(3) = 300\pi$

35 [Intermediate] • Target: Explaining when polynomial division is required

A line has vector equation $\mathbf{r} = (-1; 4; 6) + t(-1; 4; 2)$. Find the point when $t = -4$.

a. $(3, -12, -2)$ ✓

b. $(4, -10, -3)$

c. $(2, -10, 0)$

d. $(5, -14, 0)$

Working through this vectors step: $\mathbf{r} = (-1; 4; 6) + (-4)(-1; 4; 2) = (3, -12, -2)$

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36 [Intermediate] • Target: Repeated irreducible quadratic factor

An investment grows by 3%, 2%, 18% in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 8.93\%

b. 5.93\%

c. 9.43\%

d. **7.43\%** ✓

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.03 \times 1.02 \times 1.18} - 1 \approx 7.43\%$

37 [Intermediate] • Target: Verifying a decomposition by substitution

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$ ✓

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

First, note that $x = 0$ is not a solution to $e^x - kx^2 = 0$ since $e^0 = 1eq0$. Therefore, we can rewrite the equation as: $k = \frac{e^x}{x^2}$ Let $f(x) = \frac{e^x}{x^2}$ defined for all $xeq0$. The number of distinct real roots corresponds to the number of points of intersection between the curve $y = f(x)$ and the horizontal line $y = k$. Differentiating $f(x)$ using the quotient rule: $f'(x) = \frac{e^x \cdot x^2 - e^x \cdot 2x}{x^4} = \frac{e^x x(x-2)}{x^4} = \frac{e^x(x-2)}{x^3}$ Setting.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Choosing efficient substitution values • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Which of the following is the correct expression of $\frac{5x+1}{(x-1)(x+2)}$ in partial fractions?

a. $\frac{2}{x-1}+\frac{3}{x+2}$ ✓

b. $\frac{3}{x-1}+\frac{2}{x+2}$

c. $\frac{2}{x-1}-\frac{3}{x+2}$

d. $\frac{1}{x-1}+\frac{4}{x+2}$

To decompose the rational function into partial fractions, set up the identity with distinct linear factors: $\frac{5x+1}{(x-1)(x+2)} \equiv \frac{A}{x-1}+\frac{B}{x+2}$ Multiplying both sides by the common denominator $(x-1)(x+2)$: $5x+1 \equiv A(x+2)+B(x-1)$ To determine A, substitute $x=1$: $5(1)+1=A(1+2)+B(0) \Rightarrow 6=3A \Rightarrow A=2$ To determine B, substitute $x=-2$: $5(-2)+1=A(0)+B(-2-1) \Rightarrow -9=-3B \Rightarrow B=3$ Therefore, the partial.

39 [Challenging] • Target: Distinct linear factors • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

The diagram shows the graph of the function $y=f(x)$. The equation of the function is given by $f(x)=x^2-4x+3$. Identify the coordinates of the local minimum of the graph.

a. **(2, -1)** ✓

b. (1, 0)

c. (3, 0)

d. (2, 1)

The function is given by $f(x)=x^2-4x+3$. To find the local minimum, we first find the derivative of the function: $f'(x)=2x-4$. We set the derivative to zero to find the critical points: $2x-4=0$, which gives $x=2$. To confirm this is a minimum, we can find the second derivative: $f''(x)=2$. Since $f''(2)=2>0$, the critical point at $x=2$ corresponds to a local minimum. Now we find the y-coordinate by substituting $x=2$ into the original function.

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40 [Challenging] • Target: Repeated linear factors • Cross-Lesson: 3.01 3D Vectors & the Cross Product

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the x-coordinate of the vertex of the parabola.

a. 1

b. **2** ✓

c. 3

d. 4

The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Alternatively, the x-coordinate of the vertex is the midpoint of the roots. The roots are the x-intercepts, which can be found by setting $f(x) = 0$: $x^2 - 4x + 3 = 0$. Factoring this gives $(x - 1)(x - 3) = 0$, so the roots are $x = 1$ and $x = 3$. The midpoint is.

41 [Challenging] • Target: Irreducible quadratic factors • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

The diagram shows the graph of a quadratic function $y = f(x) = x^2 - 6x + k$. The graph intersects the x-axis at point A, and has a minimum point at B. The coordinates of point A are $(3, 0)$. Find the value of k.

a. -9

b. 0

c. **9** ✓

d. 18

The function is given by $f(x) = x^2 - 6x + k$. We are given that the graph intersects the x-axis at point A $(3, 0)$. This means that when $x = 3$, $f(x) = 0$. Substituting these values into the function's equation: $0 = (3)^2 - 6(3) + k\ 0 = 9 - 18 + k\ 0 = -9 + k\ k = 9$ Therefore, the value of k is 9. Common algebraic traps include incorrectly substituting the coordinates or making errors in the order of operations when solving for k. For example, mistaking the point $(3, 0)$ as the vertex would lead to an in.

42 [Challenging] • Target: Applying partial fractions to integration • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Consider the graph of the quadratic function $y = x^2 - 4x + 3$. What are the coordinates of the vertex of this parabola?

a. **(2, -1)** ✓

b. (1, 0)

c. (3, 0)

d. (2, 1)

The given quadratic function is in the form $y = ax^2 + bx + c$, where $a = 1, b = -4, c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -b/(2a)$. In this case, $x = -(-4)/(2 \times 1) = 4/2 = 2$. To find the y-coordinate of the vertex, substitute this x-value back into the function: $y = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (2, -1).

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43 [Challenging] • Target: Applying partial fractions to a binomial series expansion • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

$\int_0^{\infty} \frac{1}{1+x^4}\,dx$: evaluate using partial fractions over $\mathbb{R}$:

a. $\pi/2$

b. $\pi/4$

c. $\pi/(2\sqrt{2})$ ✓

d. π

Applying Techniques of Integration principles, $= \pi/(2\sqrt{2})$. Therefore, the verified correct option is $\pi/(2\sqrt{2})$.

44 [Challenging] • Target: Combining terms before decomposing • Cross-Lesson: 4.02 The Poisson Distribution

Approximate $\int_0^1 \frac{\sin x}{x}dx$ using power series (first three terms):

a. $1 - 1/3 + 1/10$

b. $1 - 1/18 + 1/600$ ✓

c. $1 - 1/6 + 1/20$

d. $1 - 1/18$

$\sin x/x = \sum(-1)^n x^{2n}/(2n+1)!$; $\int_0^1 = \sum(-1)^n/(2n+1)(2n+1)! = 1 - 1/18 + 1/600 - \cdots$.

45 [Challenging] • Target: Applying partial fractions to a rate problem • Cross-Lesson: 5.01 Further Integration Techniques

Find the constant term of the quotient when $4x^2 + 6x + 3$ is divided by $(x + 1)$.

a. **2** ✓

b. 1

c. 4

d. 3

Dividing: quotient $= 4x + 2$, remainder $= 1$. Constant term of quotient: 2.

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